

Practice Test — Chapter 2: Genetics

Year 10 Science (Victorian Curriculum, VCSSU119)

Name: _____ Date: _____

Time allowed: 50 minutes Total marks: /50

Instructions: Answer all questions. Show your working for genetics crosses. The answer key is at the end — try the whole test before checking.

Section A — Multiple choice (10 marks, 1 mark each)

Circle the best answer.

1. The basic building block (subunit) of DNA is called a: - A. chromosome - B. nucleotide - C. gene - D. ribosome
 2. In DNA, adenine always pairs with: - A. guanine - B. cytosine - C. thymine - D. uracil
 3. Which base is found in RNA but **not** in DNA? - A. adenine - B. thymine - C. guanine - D. uracil
 4. A human body (somatic) cell contains how many chromosomes? - A. 23 - B. 24 - C. 46 - D. 92
 5. The process that produces four genetically unique haploid gametes is: - A. mitosis - B. meiosis - C. fertilisation - D. transcription
 6. An organism with the genotype **Tt** is described as: - A. homozygous dominant - B. homozygous recessive - C. heterozygous - D. codominant
 7. Which of these is an example of an X-linked recessive condition? - A. Down syndrome - B. haemophilia - C. Huntington's disease - D. sickle cell anaemia
 8. A mutation that occurs in a body (somatic) cell: - A. is always beneficial - B. is passed on to all offspring - C. is not passed on to offspring - D. always causes cancer
 9. Down syndrome (trisomy 21) is caused by: - A. a deletion of chromosome 21 - B. three copies of chromosome 21 - C. a point mutation in chromosome 21 - D. a missing X chromosome
 10. AB blood type is an example of: - A. complete dominance - B. incomplete dominance - C. codominance - D. a recessive trait
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Section B — Short answer (16 marks)

11. Describe the relationship between **DNA, genes and chromosomes**. (3 marks)
12. Name the three parts of a nucleotide. (3 marks)
13. Define each term: (4 marks)
 - a) genotype

- b) phenotype
- c) homozygous
- d) heterozygous

14. Compare **mitosis** and **meiosis** by stating two key differences. (2 marks)

15. Explain why the **father** determines the sex of a child. (2 marks)

16. Define the term **mutation** and explain the difference between a **spontaneous** and an **induced** mutation. (2 marks)

Section C — Application and problem solving (16 marks)

17. A strand of DNA has the following base sequence: (4 marks)

T A C G G A T C A

- a) Write the complementary DNA strand. (2 marks)
- b) Write the mRNA strand transcribed from the original strand. (2 marks)

18. In pea plants, tall (T) is dominant to dwarf (t). Two heterozygous tall plants (Tt) are crossed. (6 marks)

- a) Complete a Punnett square for this cross. (3 marks)
- b) State the **genotype ratio** of the offspring. (1 mark)
- c) State the **phenotype ratio** of the offspring. (1 mark)
- d) What is the chance (%) of producing a dwarf plant? (1 mark)

19. In humans, brown eyes (B) are dominant to blue eyes (b). A heterozygous brown-eyed man has children with a blue-eyed woman. (4 marks)

- a) State the genotypes of both parents. (2 marks)
- b) Draw a Punnett square and state the percentage chance of having a blue-eyed child. (2 marks)

20. Name the type of point mutation shown in each example below: (2 marks)

- a) ATG → ACG
 - b) ATG → AG
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Section D — Extended response (8 marks)

21. Explain how a gene is used to make a protein. In your answer, refer to **transcription**, **translation**, the role of **mRNA**, **codons**, and where each process occurs. (5 marks)

22. Sickle cell anaemia is caused by a mutation in a gene that codes for haemoglobin. It is described as **both deleterious and beneficial**. Explain how a single mutation can be harmful in some situations but helpful in others. (3 marks)

ANSWER KEY — Genetics

Section A 1. B — nucleotide 2. C — thymine 3. D — uracil 4. C — 46 5. B — meiosis 6. C — heterozygous 7. B — haemophilia 8. C — is not passed on to offspring 9. B — three copies of chromosome 21 10. C — codominance

Section B

11. DNA is the genetic material; it is tightly wound (around histone proteins) to form chromosomes; genes are sections (lengths) of DNA on a chromosome that code for proteins. (3)

12. A five-carbon sugar (deoxyribose); a phosphate group; a nitrogenous base (A, T, G or C). (3)

13. a) genotype = the combination of alleles for a gene; b) phenotype = the physical appearance/expression of a trait; c) homozygous = two identical alleles; d) heterozygous = two different alleles. (4)

14. Any two: Mitosis produces 2 daughter cells, meiosis produces 4; mitosis produces identical diploid cells, meiosis produces unique haploid cells; meiosis involves crossing over/reshuffling, mitosis does not; mitosis is for growth/repair, meiosis is for making gametes. (2)

15. The father carries the sex chromosomes X and Y and can pass on either an X (→ female) or a Y (→ male). The mother can only pass on an X chromosome, so the father's contribution determines sex. (2)

16. A mutation is a change in the genetic code (DNA) of a cell. A spontaneous mutation occurs naturally; an induced mutation is caused by exposure to a mutagen (e.g. radiation or chemicals). (2)

Section C

17. a) Complementary DNA: ATG CCT AGT (2) b) mRNA: AUG CCU AGU (2) (*mRNA is complementary to the original strand, with U replacing T.*)

18. a) Punnett square (gametes T, t × T, t): (3)

	T	t
T	TT	Tt
t	Tt	tt

b) Genotype ratio = 1 TT : 2 Tt : 1 tt (1)

c) Phenotype ratio = 3 tall : 1 dwarf (1)

d) 25% chance of dwarf (1)

19. a) Man = Bb, woman = bb (2) b) Punnett square:

	b	b
B	Bb	Bb

	b	b
b	bb	bb

50% chance of a blue-eyed (bb) child. (2)

20. a) substitution; b) deletion. (2)

Section D

21. Marking points (5): The gene's base sequence holds the code. **Transcription** occurs in the **nucleus**: the DNA "unzips" and one strand is used as a template to build a complementary **mRNA** strand (with uracil replacing thymine). The mRNA leaves the nucleus and travels to a **ribosome**. **Translation** occurs at the ribosome: the mRNA is read in groups of three bases called **codons**, each specifying an amino acid (brought by tRNA). The amino acids join to form a protein. *(1 mark each, max 5.)*

22. Marking points (3): A **deleterious** (harmful) mutation reduces survival/reproduction — sickle cell causes misshapen red blood cells that block vessels. A **beneficial** mutation increases survival. In **non-malarial** regions the mutation is harmful (causes disease). In **malarial** regions, carriers have increased resistance to malaria, so it is beneficial — the same mutation has different effects depending on the environment. *(Award for: harmful effect, beneficial effect, role of environment/malaria.)*

END OF TEST